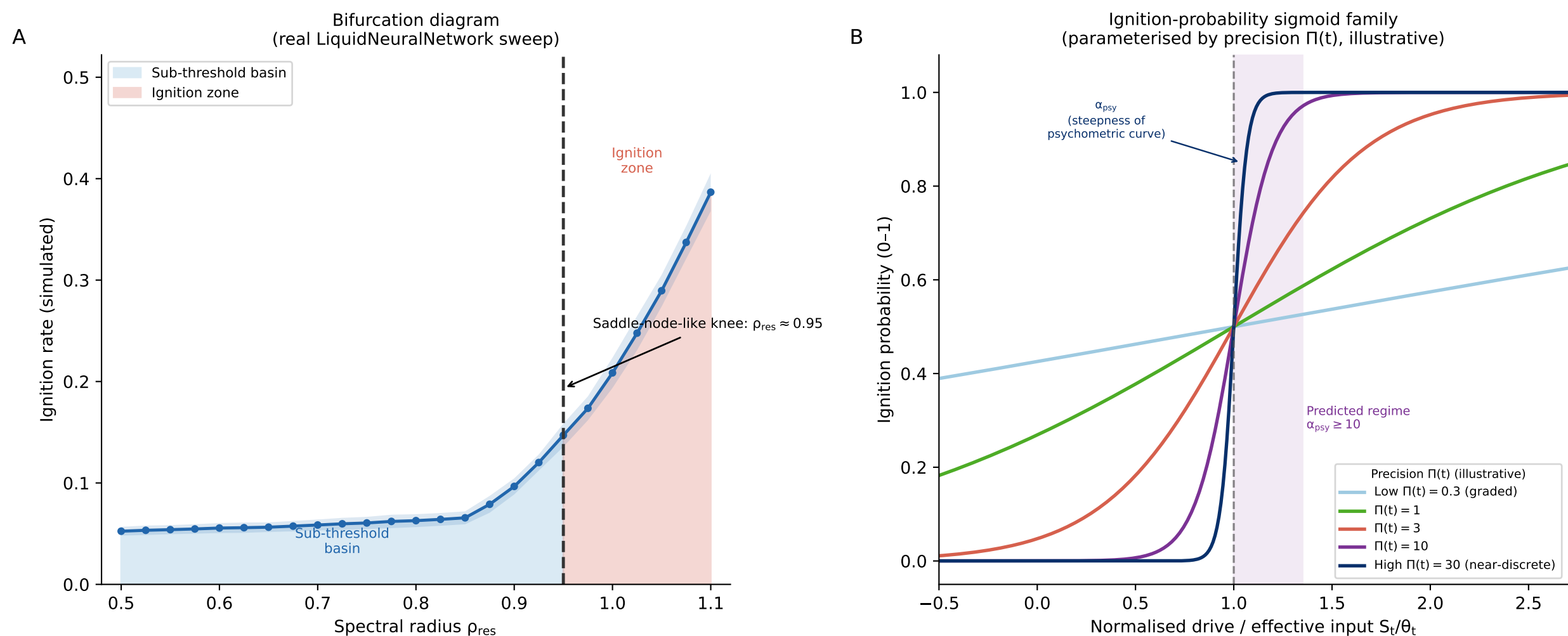




Heterogeneous- τ form used (§4.1). Panel A: real LiquidNeuralNetwork simulation, spectral radius ρ_{res} swept over [0.5, 1.1], $N=1200$ steps x 12 seeds per point; S_t proxy = $\|x(t)\|$ (reservoir-state norm) vs. $\theta_t = 2.4$, post-ignition reset $x < -\rho_S \cdot x$ ($\rho_S = 0.5$). Proof-of-concept scale only; biologically realistic validation ($N \geq 1,000$ units) pending. Panel B: γ_{sig} (model-internal steepness, $= 1/\tau_\sigma$, canonical range [2, 7.5]) and α_{psy} (behavioural psychometric steepness, predicted ≥ 10) are related but NOT interchangeable (§4.2); curves are parameterised by precision $\Pi(t)$, illustrative pre-data predictions.